Unit 6: NEW ENGINEERING MATERIALS

SEMICONDUCTOR MATERIALS:

- Introduction of Group IV elements,
- Properties,
- Concept of carriers,
- Concept of bands and band gap,
- P-type, N-type materials
- Introduction to P-N Junction Diode and I-V characteristics,
- Zener diode and its characteristics

Sr. no.	Questions
1.	Define Semiconductor. **Sol:** **A semiconductor is a material with electrical conductivity that falls between that of a conductor and an insulator.** This is due to the material's energy band structure, which allows it to conduct electricity under certain conditions. **Key characteristics:** • **Energy gap:** Semiconductors have a small energy gap (typically 1-3 eV) between the valence band (where electrons are bound) and the conduction band (where electrons are free to flow). • **Carrier concentration:** Semiconductors have a limited number of charge carriers (electrons and holes) available for conduction. • **Conductivity:** Semiconductors' conductivity can be controlled by introducing impurities (doping) or applying external stimuli (temperature, light, voltage). **Examples of semiconductor materials include:** • Silicon (Si) • Germanium (Ge) • Gallium Arsenide (GaAs) • Indium Phosphide (InP)

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Types semiconductor materials include:

Two Types:

- Single Crystal Semiconductors
- Compound Semiconductors

Single Crystal Semiconductors:

- **Definition:** A single crystal semiconductor is a material with a continuous, uninterrupted crystal lattice structure.
- **Examples:** Silicon (Si), Germanium (Ge)

Compound Semiconductors:

- **Definition:** A compound semiconductor is a material composed of two or more elements, often with a crystalline structure.
- **Examples:** Gallium Arsenide (GaAs), Gallium Nitride (GaN), Silicon Carbide (SiC)

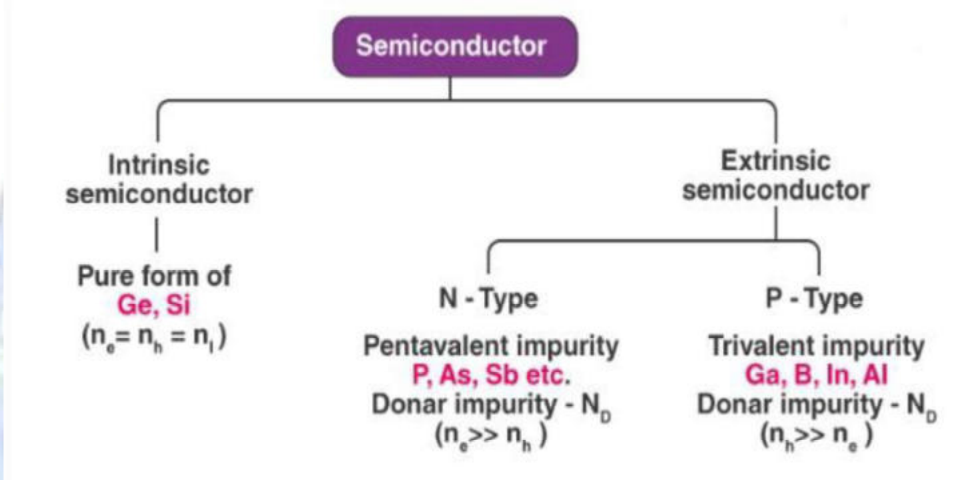
2. Classify types of semiconductor.

Sol:

Types of Semiconductors

Semiconductors can be classified as follows:

- Intrinsic Semiconductor
- Extrinsic Semiconductor



Classification of Semiconductors

3. Write Group IV element uses in detail.

Sol:

Here are some of the occurrences of Group 4 elements in nature and their uses:

Carbon (C)

- **Organic Chemistry & Life:** Carbon is the backbone of all organic molecules, making it essential for life. It forms countless organic compounds, including DNA, proteins, carbohydrates, and fats.
- **Energy:** Used in fossil fuels like coal, oil, and natural gas, which are critical energy sources. Carbon is also found in biofuels.
- **Materials:** Carbon is used to produce materials like graphite (in pencils and lubricants), diamonds (in cutting tools and jewellery), and carbon fiber (in lightweight, high-strength materials).
- **Industrial Gases:** Carbon dioxide (CO_2) is used in beverages, refrigeration, and fire extinguishers.
- **Steelmaking:** Carbon is a key component in steel production, where it is used to convert iron into steel.

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2. Silicon (Si)

- **Electronics:** Silicon is the primary material used in semiconductors and microelectronics, including computer chips, transistors, and diodes.
- **Solar Cells:** Silicon is the most common material used in photovoltaic cells for solar energy.
- **Construction:** Silicon dioxide (sand, quartz) is a major component in glass, concrete, and ceramics.

3. Germanium (Ge)

- **Optics:** Germanium is used in infrared optics, including lenses and windows for infrared cameras and thermal imaging devices.
- **Semiconductors:** Used in semiconductor devices like transistors and diodes, often in high-speed electronics and fiber-optic communication systems.
- **Solar Cells:** Germanium is used as a substrate in high-efficiency solar cells, particularly in space applications.

4. Tin (Sn)

- **Solder:** Tin is widely used in solder, which is essential for joining electrical and electronic components.
- **Coatings:** Tin is used to coat other metals (like steel) to prevent corrosion, such as in tin cans used for food packaging.

5. Lead (Pb)

- **Batteries:** Lead is a primary component in lead-acid batteries, commonly used in cars and backup power systems.
- **Radiation Shielding:** Due to its high density, lead is used in shielding against radiation in medical and industrial applications.
- **Ammunition:** Lead is used in bullets and shot.

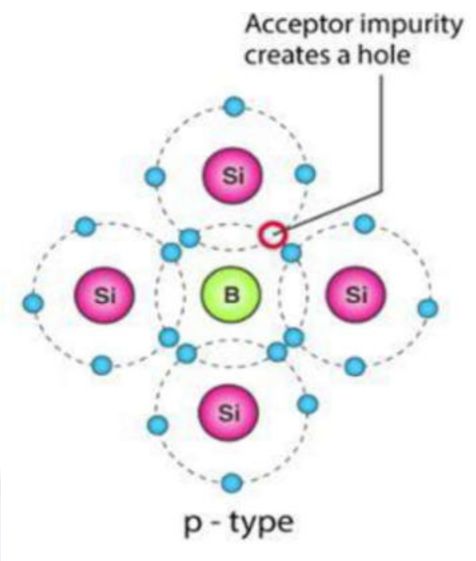
4. Difference Between / Compare intrinsic & extrinsic semiconductor.

Sol:

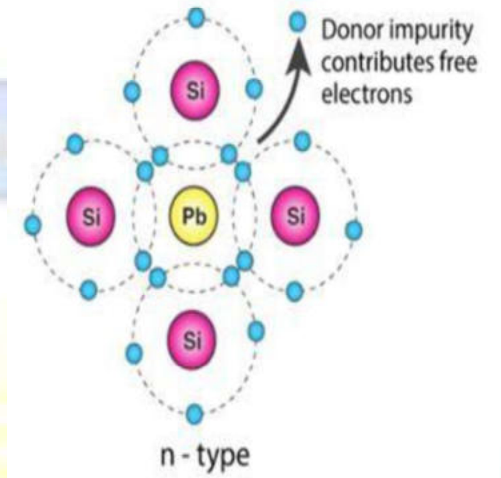
Intrinsic Semiconductor	Extrinsic Semiconductor
Pure semiconductor	Impure semiconductor
The density of electrons is equal to the density of holes	The density of electrons is not equal to the density of holes
Electrical conductivity is low	Electrical conductivity is high
Dependence on temperature only	Dependence on temperature, as well as on the amount of impurity
No impurities	Trivalent impurity and pentavalent impurity

5.	Difference Between p-type and n-type semiconductors.

Sol:



(a)



(b)

P Type Semiconductor	N Type Semiconductor
In this type of semiconductor trivalent impurities like Boron, Aluminium, Gallium are added.	In this type of semiconductor pentavalent impurities like Phosphorus, Arsenic are added.
Holes are majority carriers and electrons are minority carriers.	Electrons are majority carriers and holes are minority carriers.
Density of holes is much greater than density of electrons i.e $n_h > n_e$.	Density of electrons is much greater than density of holes i.e $n_e > n_h$.
The acceptor energy level is close to valence band and away from conduction band.	The donor energy level is close to conduction band and away from valence band.
Fermi level lies between acceptor level and valence band.	Fermi level lies between donor level and conduction band.
Impurity atom creates a vacancy of electrons i.e holes.	Impurity atom provides an extra electron.

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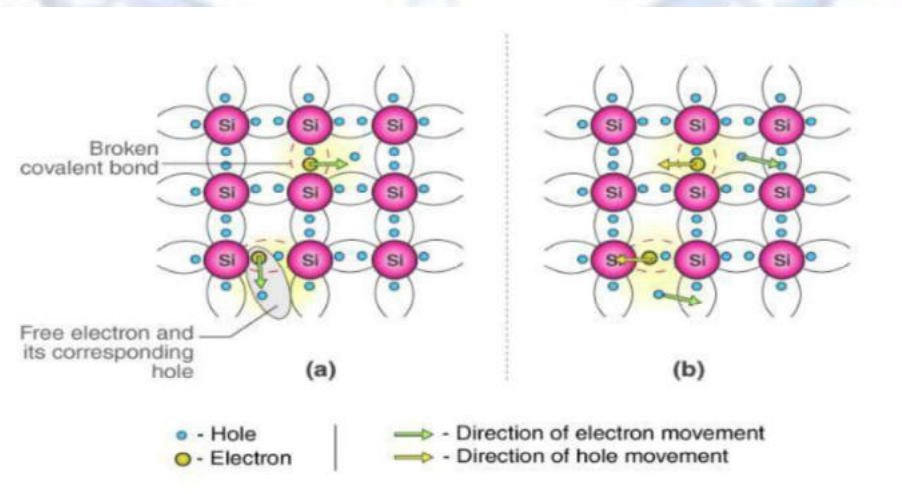
6.	Define Conduction band, Valance band and Forbidden/Energy gap. **Sol:** **Conduction band** • Formed by grouping the energy levels of free electrons • Generally, the conduction band is empty • But, when external energy is applied the electrons in the valence band jumps into the conduction band. **Valance band** • Formed by grouping the energy levels of outmost orbit electrons • Electrons in valence band are loosely bound to the nucleus of atoms. • They can be moved to conduction band by applying energy • Electrons in valence band have lower energy than conduction band **Forbidden/Energy gap** • Energy gap present between the valence band and conduction band Separating two energy bands is called forbidden gap. • Electrons cannot stay in forbidden gap because there is no allowed energy state in this region.
7.	Explain intrinsic and extrinsic (N & P type) semiconductors with the help of energy band diagram. **Sol: Intrinsic Semiconductor**  Conduction Mechanism in Case of Intrinsic Semiconductors (a) In the absence of an electric field (b) In the presence of an electric field

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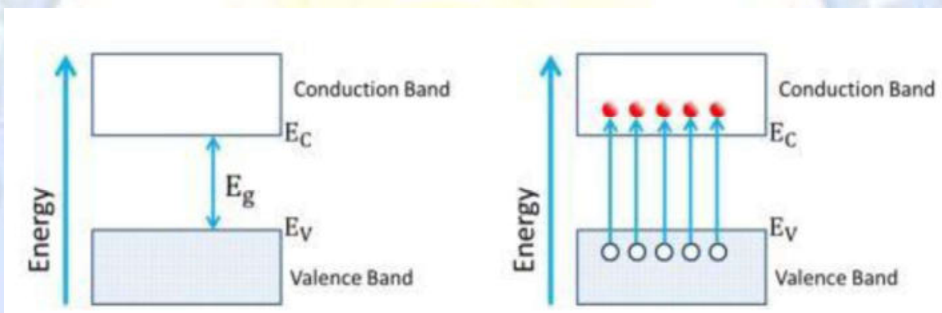
- Intrinsic semiconductor is one that is made of semiconductor material in its extremely pure form.
- Pure Silicon and pure Germanium are examples of intrinsic semiconductors.
- At absolute zero temperature, the electrons in the outer shell of intrinsic semiconductor are tightly bound with the covalent bonds.
- Hence, there are no free electrons and it behaves as insulator. However, at room temperature, some valence electrons absorb enough energy to enable them to break free from their covalent bonds.
- When this occurs, the liberated electrons are free to move about in the crystal.
- When battery is connected to this crystal, free electrons are attracted to the positive terminal of the battery and constitute electric current.
- Greater the temperature, greater is the current.
- Free electron means that its motion is quite sensitive to applied electric field.
- Generation of free electrons can also be explained on the basis of band theory.

The energy band diagram of intrinsic semiconductor is as shown in following fig. –



- In intrinsic semiconductor, at 0K, the valence band is completely filled and conduction band is completely empty.
- These bands are separated by energy gap Eg.
- In valence band, electrons are present but vacant energy states are not there so that they can move freely.
- In conduction band, vacant energy states are present but electrons are not there to move through these states freely.
- However, as the temperature increases, some of the valence band electrons get enough energy to overcome the energy gap and jump to the conductions band.
- Now, we have some vacant energy states in the valence band, which are called, as „holes‟ and number of electrons, which can move freely through holes when electric field is applied. Also, in conduction band, we have some electrons, which can move, freely through already available vacant states when electric field is applied.
- Thus, in intrinsic semiconductor, electrons and holes are generated in pairs.
- At any temperature, the number of holes valence band and the number of electrons in conduction band are same.
- As the temperature increases, the number of electron hole pairs increase.
- Hence the conductivity of the intrinsic semiconductor is directly proportional to the temperature.

Extrinsic Semiconductor

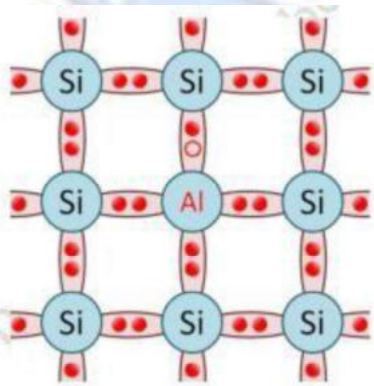
- Doping means adding impurity atoms to the crystal to increase either the number of free electrons or the number of holes.
- When the crystal is doped, it is called as an extrinsic semiconductor.
- There are two types of extrinsic semiconductors depending on the type of impurity (doping material).

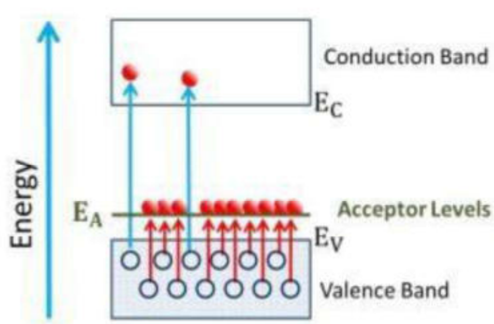
An extrinsic semiconductor can be further classified into types:

- **P-type Semiconductor**
- **N-type Semiconductor**

P – type semiconductors:

- It is formed by doping a trivalent impurity in Si or Ge. e.g. Ga, In, B.
- Valence band electrons can move freely through the holes when electric field is applied.
- Trivalent atoms accept one electron from a Silicon atom and in a process causes the semiconductor atom to have a hole.
- Hence, they are known as acceptor atoms.
- From the energy band structure viewpoint, the acceptor atoms introduce available energy states called acceptor levels just above the valence band.
- The electrons from valence band can easily jump to these acceptor levels leaving hole in the valence band.
- In p-type semiconductor, the number of holes in valence band is much more than the number conduction band electrons.
- Hence, the current is mainly due to holes.
- Thus, in p-type semiconductor, holes are majority carriers and electrons are minority carriers.





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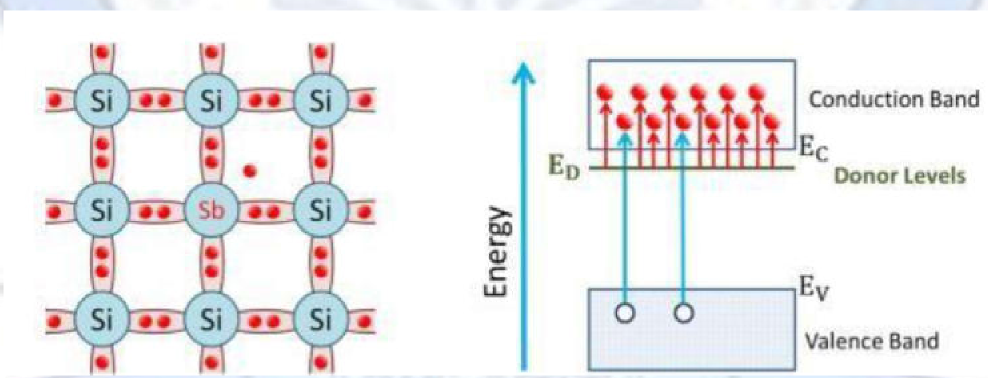
N-Type Semiconductor

- In n-type semiconductor, a pentavalent impurity such as Antimony (Sb), Arsenic (As), Phosphorous (P) is added to pure semiconductor to get extra conduction band electrons.
- The pentavalent atom (Sb) has five electrons in its valence orbit, out of which four electrons form covalent bonds with four neighbouring semiconductor (Si) atoms and fifth electron remains loosely bound as shown below-
- This loosely bound electron can be excited to conduction band easily and can be made free.
- As pentavalent atoms donate one electron to the conduction band of pure silicon, they are called as donor atoms.
- From the energy band structure viewpoint, the donor atoms introduce filled states called „donor levels‟ just below the conduction band.
- The donor levels have energy approximately 0.1 eV below the lowest level in the conduction band.
- Hence, donor electrons in donor levels can be easily excited to the conduction band.
- Once they come in conduction band, they can move freely when electric field is applied.
- This number of donated electrons in conduction band is much more as compared to the number of electron hole pairs.
- Hence, in n-type semiconductor, the electrons are majority carriers and holes are minority carriers.



8. Explain P-N junction diode in forward and reverse biasing with the help of proper diagram and its construction, working and I-V characteristics.

Sol:

What is P-N Junction?

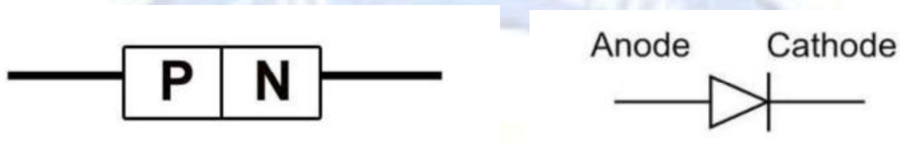
Definition: **A P-N junction is an interface or a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.**

P-N Junction Diode:

Two Operating Regions:

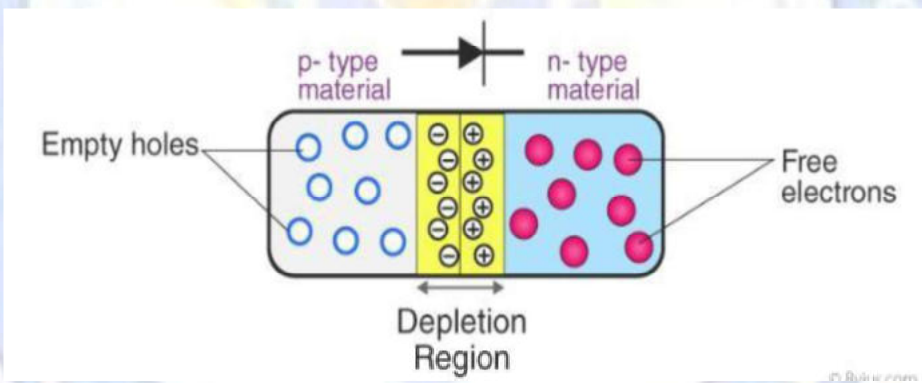
- P-type (Positive)
- N-type (Negative)

Symbol:



(a)

PN Junction Diode



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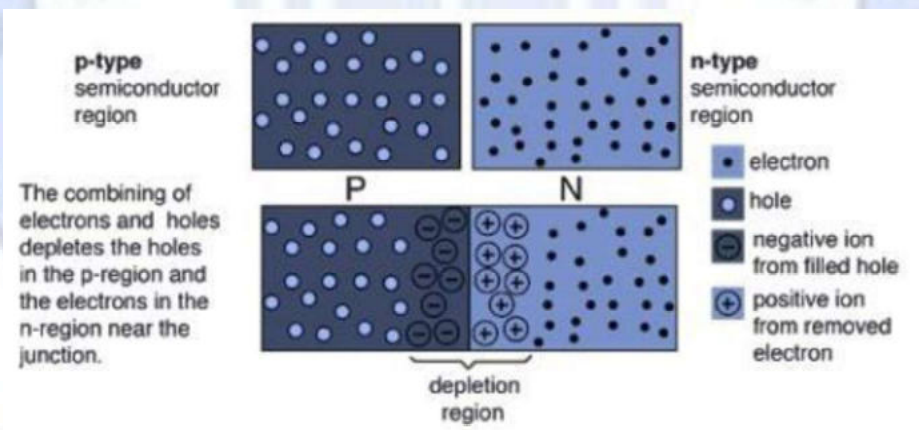
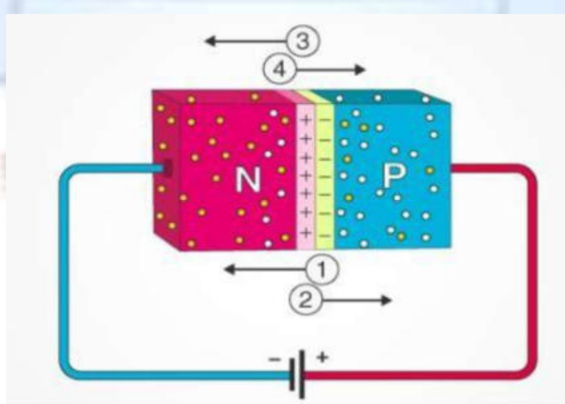
Biasing Conditions:

Zero Bias:

- P type semiconductor has Holes as majority charge carriers.
- N type semiconductor has Electron as majority charge carriers.
- Majority charge carriers near or at the junction diffuse across the junction and recombines.
- After few combination process stops.
- As electrons crossing the junction are repelled by negative ions and holes are repelled by positive ions created.
- The immobile ions at the junction creates a zone that is devoid of charge carrier is known as **"Depletion Region".**
- Thickness of Depletion region is of an order of 10^{-6} m.
- The potential difference across the depletion region is called "Potential Barrier".
- Potential Barrier can be increased or decreased by applying Voltage.



Forward Bias:



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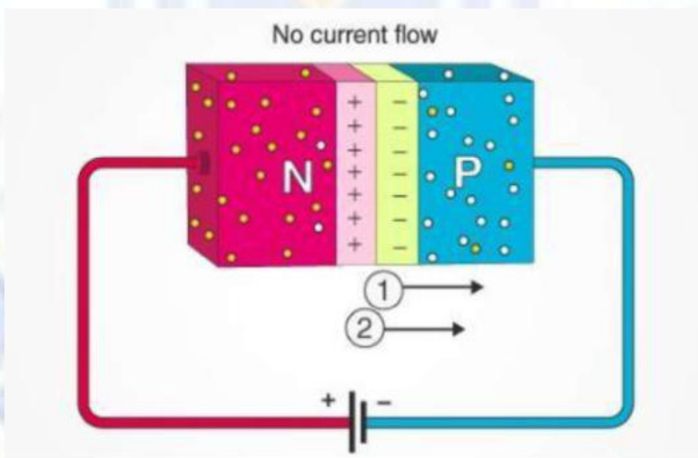
- Positive voltage on P-type
- Negative voltage on N-type
- Current flows easily

Forward Biasing a P-N Junction:

- Connect p-type to battery's positive terminal and n-type to negative terminal.
- Built-in electric field and applied electric field are in opposite directions.
- Resultant electric field has a lesser magnitude than built-in electric field.
- Depletion region becomes less resistive and thinner.
- Resistance of depletion region becomes negligible at high applied voltage.
- In silicon, at 0.6 V, depletion region's resistance becomes negligible.
- Current flows across depletion region unrestricted.

<u>Reverse Bias:</u>



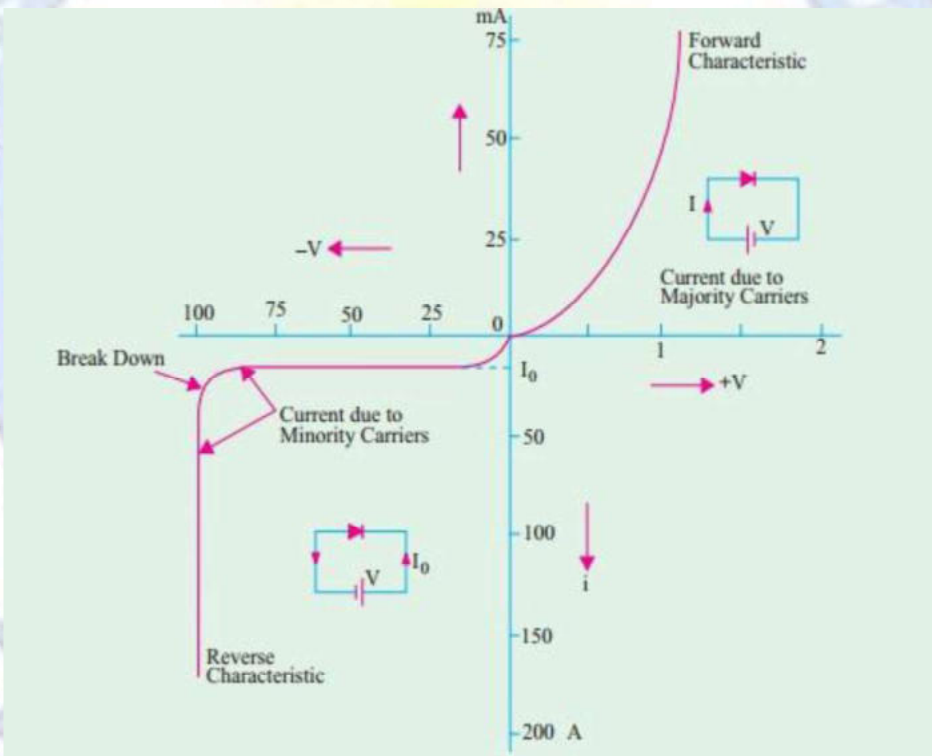
- Negative voltage on P-type
- Positive voltage on N-type
- Current blocked

Reverse Biasing a P-N Junction:

- Connect p-type to battery's negative terminal and n-type to positive terminal.
- Built-in electric field and applied electric field are in the same direction.
- Resultant electric field is stronger than built-in electric field.
- Depletion region becomes more resistive and thicker.
- Increasing applied voltage makes depletion region even more resistive and thicker.
- Current is blocked by the depletion region.

V/I Characteristics

In the fig, low powered PN junction diode's voltages/ current characteristics have been shown.



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Forward Characteristics:

1. When a diode is forward-biased, it initially blocks current flow.
2. As the applied voltage increases, the barrier voltage (0.7V for silicon, 0.3V for germanium) is overcome.
3. Once the barrier is overcome, current flows quickly with a small increase in voltage.
4. A small voltage increase (1V) can generate a large current (50mA).
5. Exceeding the safe limit of forward voltage can damage the diode.

Reverse Characteristics:

1. When a diode is reverse-biased, it blocks majority carriers and allows minimal current flow (leakage current).
2. The leakage current is very small (Nano-amperes for silicon, microamperes for germanium).

The leakage current is independent of reverse voltage but depends on:

- Temperature
- Doping level
- Physical size of the junction

3. If the reverse voltage exceeds a certain limit (breakdown voltage), the leakage current increases sharply.
4. Exceeding the breakdown voltage can damage the diode, so a current-limiting resistor is used to safeguard it.

Applications

1. P-N junction diodes are used in clamping circuits for dc restoration.
2. Used in clamping circuits for wave shaping.
3. Used in voltage multipliers.
4. Used as switch in digital logic circuits.
5. Used in demodulation circuits and optical communications.

9. What is Zener diode? Explain with circuit diagram how a Zener diode operates in reverse bias condition.

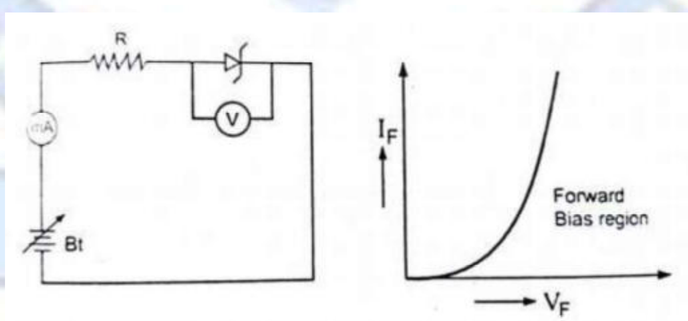
Sol:

Zener Diode (is a type of P – N Junction)

- It is named after Clarence Zener.
- A Zener diode is a type of P-N Junction diode that allows current flow not only from its anode to cathode : but also in reverse direction when the Zener voltage is achieved.
- The operation depends upon the heavy doping of p-n junction.
- Depletion layer : <1 μm
- Electric filed is about 500 KV/m.
- Small reverse bias voltage allows the electrons to tunnel from valence band of p-type material to the conduction band of n-type material.
- Designed to operate in the breakdown region without damage.

Working of a Zener Diode

- When a Zener diode is connected in forward bias, it will act like a normal p-n junction diode, having a voltage drop of around 0.7 V.



(a) Forward Bias

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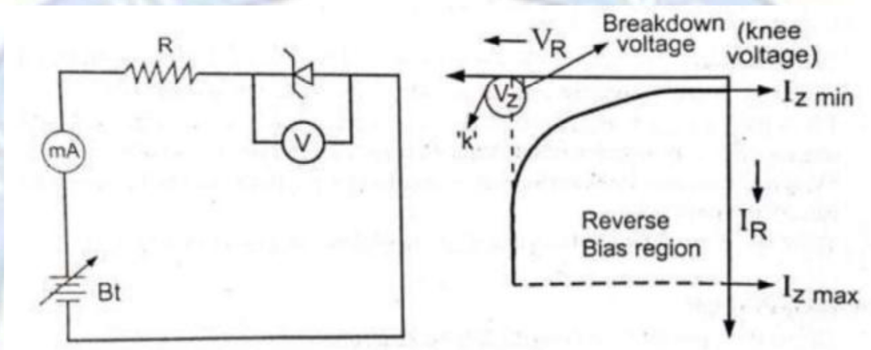
- When a zener diode is connected in reverse bias, the breakdown of a Zener diode occurs.
- Reverse current stays negligible upto point K-Knee point.
- The voltage is very sharp compared to normal p-n junction.
- Reverse current increases sharply.
- The external resistance connected to the circuit prevents the Zener diode from burning.



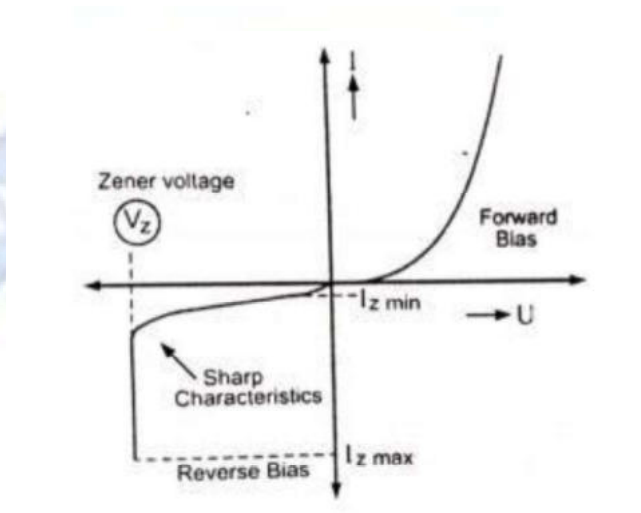
(b) Reversed Bias

- It is named after Clarence Zener who discovered Zener effect.
- "A Zener diode is a type of p-n junction diode that allows current flow not only from its anode to cathode, but also in reverse direction when Zener voltage is achieved."
- The Zener diode's operation depends on the heavy doping of its p-n junction.
- The Zener diode is designed to operate in the break down region without damage

Zener Breakdown/Zener Effect: (both words are same effect or breakdown)

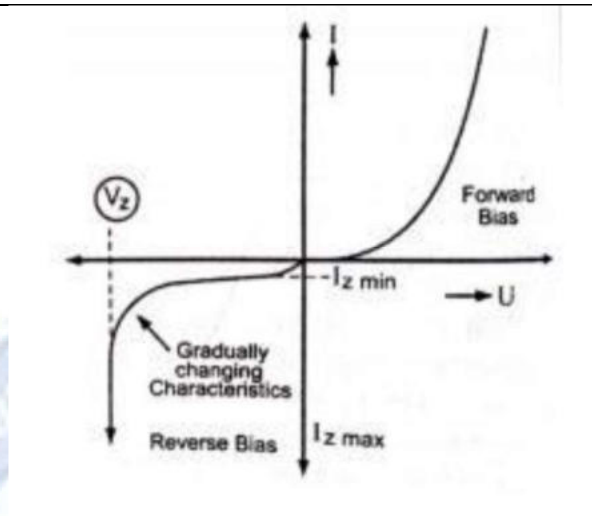
- When a reverse voltage is applied to a Zener diode it causes a very high electric field to appear across a narrow depletion layer.
- Strong electric field breaks the covalent bond of valence electrons and make them jump for conduction as free electrons.

- A large number of free electrons will constitute a large reverse current through Zener diode and breakdown is said to have occurred due to Zener effect.
- A current limiting resistance connected to Zener diode protects it against the damage due to overheating.



Avalanche Breakdown

- It reverses the bias condition of a Zener diode , the conduction takes place only due to the minority carriers.
- As reverse voltage is applied to the Zener diode , these minority carriers tend to get accelerated.
- The kinetic energy increases.
- The accelerated minority carriers collide with the stationary atoms and impart some of the kinetic energy to the valence electrons present in the covalent bonds.
- The additional acquired energy make electrons break their covalent bonds and jump into conduction band to become free from conduction.
- The newly generated free electrons will get accelerated and by means of collisions, will generate carrier multiplication, which leads to the breakdown called Avalanche breakdown.



Application:

- Highly used as a voltage regulator
- Used in wave shaping circuits as peak limiters or clippers
- Used for meter protection to prevent against damage from accidental overloads.

10. Difference between Zener breakdown and Avalanche breakdown.

Sol:

Avalanche Breakdown	Zener Breakdown
It occurs in junctions which are lightly and have wide depletion widths.	It occurs in junctions which are heavily doped and have narrow depletion widths.
It occurs at higher reverse voltages when thermally generated electrons get enough kinetic energy to produce more electrons by collision.	It occurs due to rupture of covalent bonds by strong electric fields set up in depletion region by the reverse voltage.
At reverse voltage above 6V breakdown is due to avalanche effect.	At reverse voltage below 6V breakdown is due to zener effect.
Electric field produced is weak in nature.	A strong electric field is produced
Charge carriers obtain energy from the applied potential.	Zener current is independent of applied voltage.

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